

GLASGOW, KY, USA

WMO: 722322

Lat: 37.033N

Lon: 85.950W

Elev: 716

StdP: 14.32

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 00361

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.2	16.3	-3.7	4.6	17.7	1.4	6.1	19.0	21.0	49.1	19.1	49.2	3.0	350	0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	93.2	75.2	91.1	74.4	89.9	74.1	78.5	87.4	77.4	86.7	76.3	84.9	5.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.6	142.7	81.5	75.0	134.7	80.3	73.4	127.7	79.4	42.6	87.5	41.2	86.6	40.1	85.1	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.7	15.4	12.6	3.0	96.9	9.2	4.2	-3.7	100.0	-9.0	102.5	-14.2	104.8	-20.9	107.9		
			1.6	80.3	8.4	2.3	-4.4	82.0	-9.3	83.3	-14.0	84.6	-20.1	86.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.1	36.2	39.3	48.4	58.0	67.4	75.2	78.0	76.8	71.1	58.9	46.8	40.1
	DBStd	17.31	11.74	11.44	11.17	9.37	7.59	4.75	4.37	4.55	6.44	9.31	9.59	10.38
	HDD50	1519	450	330	170	33	1	0	0	0	0	24	171	340
	HDD65	4030	893	721	523	243	70	2	0	0	22	235	550	771
	CDD50	4481	22	30	119	274	539	756	868	831	632	301	75	34
	CDD65	1513	0	0	7	33	143	308	403	366	204	46	3	0
	CDH74	14091	0	1	56	304	1240	2887	3827	3331	1949	479	17	0
	CDH80	5654	0	0	4	42	363	1166	1697	1440	808	132	2	0
Wind	WSAvg	4.4	5.8	6.1	6.1	5.6	3.8	3.4	3.1	2.6	2.7	3.7	4.6	5.2
Precipitation	PrecAvg	50.9	4.4	3.8	5.0	4.3	5.4	4.4	4.6	3.2	3.7	3.6	3.6	4.8
	PrecMax	60.9	7.9	6.9	13.8	7.2	12.4	9.2	8.5	7.9	8.1	7.4	5.8	11.1
	PrecMin	38.2	2.1	1.2	2.6	1.8	1.8	1.7	0.9	0.6	0.8	0.4	0.8	2.2
	PrecStd	6.6	1.3	1.4	2.6	1.6	2.6	1.9	1.7	2.2	2.0	1.3	2.0	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.2	70.5	80.7	84.0	90.1	93.5	97.0	97.1	94.9	88.3	75.7	69.7
		MCWB	55.7	59.6	63.5	66.2	71.8	74.3	74.1	79.5	72.6	73.5	61.9	60.5
	2%	DB	63.3	66.1	75.0	80.9	86.4	91.1	92.9	93.0	91.2	83.6	71.8	65.7
		MCWB	55.0	57.4	61.1	64.5	70.0	73.1	76.1	76.2	72.6	68.9	59.7	59.7
	5%	DB	60.2	62.7	71.5	77.5	84.0	89.8	90.7	90.4	88.5	79.4	67.4	61.4
		MCWB	52.9	54.6	59.1	62.1	68.7	72.9	75.5	75.1	72.3	65.5	57.9	55.7
	10%	DB	55.3	57.6	66.3	74.2	81.6	87.1	88.9	88.1	84.6	75.0	63.4	57.2
		MCWB	48.2	50.5	55.9	61.0	67.9	72.6	74.9	74.3	70.5	63.7	55.1	52.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.3	63.0	64.8	69.0	74.0	78.3	80.5	81.8	78.2	76.0	65.5	64.2
		MCDB	63.4	66.7	75.0	79.8	84.0	88.2	85.8	92.4	87.5	82.1	70.1	67.3
	2%	WB	56.8	59.6	63.2	67.1	72.2	76.7	78.3	79.2	75.2	71.8	63.0	60.8
		MCDB	61.7	65.3	72.4	76.5	82.3	86.7	86.5	88.9	84.3	79.4	69.0	64.3
	5%	WB	54.0	56.3	60.8	65.1	70.9	75.2	77.3	77.3	73.6	68.6	59.5	57.3
		MCDB	59.2	60.4	68.9	73.8	80.5	84.8	86.8	86.1	83.7	75.3	64.3	61.1
	10%	WB	49.5	52.0	58.4	63.1	69.3	73.8	76.4	75.9	72.2	65.4	56.3	53.0
		MCDB	56.1	57.5	65.1	71.0	78.4	82.9	85.2	83.5	81.7	73.6	62.1	57.4
Mean Daily Temperature Range	5% DB	MDBR	17.5	18.4	20.6	22.6	21.5	20.8	19.6	20.6	22.8	23.1	21.3	17.0
		MCDBR	20.2	22.5	24.5	25.7	24.1	23.7	22.1	23.5	26.9	26.9	25.0	20.4
		MCWBR	16.1	16.9	14.5	13.4	10.1	9.3	8.1	9.8	10.3	13.0	15.6	16.1
	5% WB	MCDBR	18.9	19.6	21.2	20.8	20.3	20.0	19.2	20.2	22.8	22.9	19.6	18.3
MCWBR		16.9	17.3	13.9	12.6	9.5	9.2	7.9	8.9	10.0	12.5	15.2	16.3	
Clear-Sky Solar Irradiance	taub	0.313	0.322	0.360	0.402	0.446	0.467	0.499	0.508	0.421	0.366	0.331	0.308	
	taud	2.470	2.431	2.371	2.251	2.174	2.149	2.079	2.040	2.265	2.409	2.466	2.492	
	Ebn at Noon	278	290	286	279	267	261	251	246	263	270	269	272	
	Edh at Noon	26	31	36	43	47	48	51	52	40	31	26	24	
All-Sky Solar Radiation	RadAvg	655	852	1199	1590	1774	1974	1874	1746	1458	1115	786	565	
	RadStd	67	114	115	116	160	114	108	92	134	130	82	55	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+234	+151

Nomenclature: See separate page